PV- EVOTWIN A SELF-EVOLVING MULTI-PHYSICS DIGITAL TWIN FOR UTILITY-SCALE SOLAR PV FARMS

PROJECT REPORT

Submitted by

SAMIDURAI V **8115U23ME037**

In partial fulfilment for the award of the course

of

MEA1411 - PROJECT PHASE 1
IN

DEPARTMENT OF

MECHANICAL ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

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Under the Guidance of

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BONAFIDE CERTIFICATE

Certified that this project report titled **"PV - EVOTWIN A SELF-EVOLVING MULTI-PHYSICS DIGITAL TWIN FOR UTILITY-SCALE SOLAR PV FARMS"** is the Bonafide work of **SAMIDURAI V (8115U23ME037),** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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CHENNAI DECLARATION BY THE CANDIDATES

I hereby declare that the project report entitled **"PV - EVOTWIN A SELF-EVOLVING MULTI-PHYSICS DIGITAL TWIN FOR UTILITY-SCALE SOLAR PV FARMS"** is the result of the original work carried out by us. To the best of our knowledge, this work has not been submitted, either in whole or in part, to **Anna University, Chennai**, or any other university or institution for the award of any degree or diploma. This project report is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Engineering**.

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SAMIDURAI V

Place: Samayapuram

Date:

ACKNOWLEDGEMENT

It is with great pride that we express our gratitude and indebtedness to our institution for the support, "**K. RAMAKRISHNAN COLLEGE OF ENGINEERING (AUTONOMOUS**)", for providing us with the opportunity to do this project.

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I thank **Dr. H. RAMAKRISHNAN, M.E., Ph.D.**, Head of the Department, Mechanical Engineering, for his encouragement in pursuing this project. We wish to convey our profound and heartfelt gratitude to our esteemed project guide, **Mr. B. PITCHIA KRISHNAN, M.E., (Ph.D).,** Assistant Professor, Department of Mechanical Engineering, who supported us in our project phase, for his incalculable suggestions, creativity, assistance and patience, which motivated us to carry out this project.

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I wish to express our special thanks to the officials and lab technicians of our department who rendered their help during the period of the work progress.

DEPARTMENT OF MECHANICAL ENGINEERING
Program: Bachelor of Engineering

VISION & MISSION OF THE COLLEGE

VISION:

- To achieve a prominent position among the top technical institutions.

MISSION:

- To bestow standard technical education par excellence through state-of-the-art infrastructure, competent faculty and high ethical standards.
- To provide education that can be used as a tool to transform the students.

VISION & MISSION OF THE DEPARTMENT

VISION:

- To provide eminent Mechanical Engineers to the society and industries through quality technical education and research.

MISSION:

- To impart quality professional education through state-of-the-art-infrastructure and resources.
- To motivate students and faculty to undertake research and development activities in the thrust areas of mechanical engineering.
- To carryout collaborative projects with industries and academics with fullest potential.

DEPARTMENT OF MECHANICAL ENGINEERING

Program: Bachelor of Engineering

Program Educational Objectives :(PEOs)

Graduates of Mechanical Engineering shall:

1. Have Successful careers in industries or opt for higher studies and research.

2. Analyze, synthesize and design mechanical and allied products addressing societal issues.

3. Exhibit technical expertise, communication skills, team work with professionalism with ethical values.

Program Specific Outcomes (PSOs)

Students shall:

PSO1: Identify, formulate and solve engineering problems in three core streams of Mechanical Engineering, i.e., Design, Thermal and Manufacturing Engineering

PSO2: Design, develop, test and implement energy efficient systems for required Engineering applications.

PROGRAM OUTCOMES

PROGRAM OUTCOMES (PO's)	
PO 1	**Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis an interpretation of data, and synthesis of the information to provide vali conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PROGRAM'S OUTCOME MAPPING

Check	PO number	Explanation
☑	**PO1 – Engineering Knowledge**	Applied thermoelectric (Peltier) principles, heat transfer and basic electronics to develop the cooling compartment.
☑	**PO2 – Problem Analysis**	Identified the high energy use, bulk and refrigerant issues of conventional cooling and analysed a solid-state alternative.
☑	**PO3 – Design/Development of Solutions**	Designed a compact dual-mode (cooling and heating) compartment with automatic temperature control, considering safety and usability.
☑	**PO4 – Conduct Investigations**	Tested the unit and evaluated cabinet temperature, cooling time and heating performance.
☑	**PO5 – Modern Tool Usage**	Used the LM35 sensor, AT89C51 microcontroller, LCD and design and fabrication tools during project development.
☑	**PO6 – Engineer and Society**	Considered the need for small-scale, portable cooling for medicines, food and enclosed spaces, with safe electrical operation.
☑	**PO7 – Environment and Sustainability**	Avoided refrigerants and moving compressor parts to reduce environmental impact.
☑	**PO8 – Ethics**	Followed responsible engineering practices and reported project results honestly, including the limitations of the unit.
☑	**PO9 – Individual and Team Work**	Contributed to the team's design, fabrication, programming and testing activities.
☑	**PO10 – Communication**	Presented the design, working method and results through reports and presentations.
☑	**PO11 – Project Management and Finance**	Planned project tasks, selected components and considered the project budget.
☑	**PO12 – Life-long Learning**	Learned about thermoelectric cooling and embedded control and considered future improvements such as solar power and PID control.

\

SUSTAINABLE DEVELOPMENT GOAL



PRIMARY SDG's USED IN THE PROJECT



SDG 7: Affordable and Clean Energy: PV - EVOTWIN is a digital twin that estimates the degradation state of utility-scale solar PV farms from routine field data. By showing how much of a plant's output loss is caused by module ageing and with what uncertainty, it helps operators plan cleaning, repair and replacement, and so keep clean energy production reliable for longer. Its contribution depends on the quality of the field data and on the parameters the data can identify, and it does not generate energy by itself.

SECONDARY SDG's USED IN THE PROJECT



SDG 9: Industry, Innovation and Infrastructure: The project combines physical modelling, Bayesian estimation and a software stack of Python, FastAPI, ThingsBoard and Grafana into a new digital infrastructure for monitoring solar plants. Its identifiability-aware method is also documented as a provisional patent filing and a research paper, which supports innovation in energy infrastructure.

ABSTRACT

Solar photovoltaic panels are exposed to changing sunlight, temperature, dust and shading, so their actual output often differs from the expected output without the user noticing. This project presents a low-cost solar photovoltaic digital twin that monitors a small solar panel in real time and compares its measured power with the power predicted by a virtual model. An ESP32 DevKit V1 microcontroller acts as the data acquisition unit. An INA219 sensor measures the current, bus voltage and power of the panel, a resistive DC voltage sensor with a 5:1 division ratio provides an independent voltage measurement, a DHT11 sensor records the air temperature and relative humidity, and an LDR light sensor module gives the relative intensity of the sunlight falling on the panel.

The ESP32 reads all the sensors at fixed intervals, averages the analogue samples to reduce noise, applies the calibration factors, converts the readings to engineering units and sends them through its built-in Wi-Fi to a monitoring dashboard. The digital twin uses the measured light intensity and temperature as inputs to calculate the power that the panel should produce, and the dashboard displays the difference between the predicted and measured values. A persistent shortfall indicates losses such as dust, partial shading, loose connections or ageing of the cells, while range checks in the program identify sensor and wiring faults.

The system is simple to build, compact and consumes little power. Its accuracy is limited by the relative nature of the LDR reading and the tolerance of the DHT11, so it is best suited to tracking trends and detecting deviations. The design can be extended with a calibrated irradiance sensor, machine learning models, cloud storage and larger solar arrays.

Keywords: Solar photovoltaic panel, Digital twin, ESP32, INA219, DHT11, LDR sensor, IoT monitoring, Performance monitoring, Fault detection, Renewable energy.

TABLE OF CONTENT

CHAPTER NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	**v**
	SUSTAINABLE DEVELOPMENT GOALS	**x**
	TABLE OF CONTENT	**xi**
	LIST OF FIGURES	**xv**
	LIST OF TABLES	**xvi**
	LIST OF ABBREVIATIONS	**xvii**
	ABSTRACT	**xviii**
1	**INTRODUCTION**	**1**
	1.1 GENERAL INTRODUCTION	1
	1.2 NEED FOR THE PROJECT	2
	1.3 IMPORTANCE OF THE WORK	3
	1.4 OVERVIEW OF THE PROPOSED PROJECT	4
	1.5 SCOPE OF THE PROJECT	4
2	**LITERATURE REVIEW**	**5**
3	**BACKGROUND STUDY**	9
	3.1 PHOTOVOLTAIC EFFECT AND MODULE STRUCTURE	11
	3.2 EQUIVALENT-CIRCUIT (SINGLE-DIODE) MODEL	11
	3.3 EFFECT OF IRRADIANCE AND TEMPERATURE	11
	3.4 THERMAL MODEL OF THE MODULE	12
	3.5 DEGRADATION MECHANISMS AND RATES	12
	3.6 SPARSE FIELD DATA AND DATA QUALITY	13
	3.7 THE INVERSE PROBLEM AND IDENTIFIABILITY	13
	3.8 BAYESIAN ESTIMATION	13
	3.9 DIGITAL TWIN CONCEPTS	13
	3.10 SELF-EVOLVING UPDATING	13
	3.11 SOFTWARE AND DATA PLATFORM	13
4	**PROBLEM IDENTIFICATION**	**14**

CHAPTER NO	TITLE	PAGE NO
	4.1 LACK OF INSIGHT INTO DEGRADATION MECHANISMS	14
	4.2 STATIC MODELS THAT DO NOT EVOLVE	14
	4.3 SPARSE SENSING AND THE ILL-POSED INVERSE PROBLEM	14
	4.4 ABSENCE OF UNCERTAINTY INFORMATION	14
	4.5 DISCONNECTED PHYSICS AND DATA-DRIVEN METHODS	14
	4.6 PROBLEM STATEMENT	15
5	**IMPACT OF THE PROBLEM**	**16**
	5.1 LOSS OF ENERGY YIELD	16
	5.2 ECONOMIC LOSS AND INVESTMENT RISK	16
	5.3 INEFFICIENT OPERATION AND MAINTENANCE	16
	5.4 WRONG DIAGNOSIS FROM ILL-POSED ESTIMATION	17
	5.5 RELIABILITY AND GRID CONSIDERATIONS	17
	5.6 ENVIRONMENTAL AND SUSTAINABILITY IMPACT	18
6	**OBJECTIVES**	18
	6.1 PRIMARY OBJECTIVES	19
	6.2 TECHNICAL OBJECTIVES	20
7	**HARDWARE REQUIREMENTS**	**21**
	7.1 SOLAR PHOTOVOLTAIC PANEL	21
	7.2 ESP32 DEVKIT V1 MICROCONTROLLER	21
	7.3 INA219 CURRENT AND POWER SENSOR	21
	7.4 DC VOLTAGE SENSOR MODULE (0–25 V)	22
	7.5 DHT11 TEMPERATURE AND HUMIDITY SENSOR	22
	7.6 LDR LIGHT SENSOR MODULE (LM393)	22
	7.7 INTERCONNECTION OF THE HARDWARE	23
8	**WORKING PRINCIPLE**	30
	8.1 OPERATING PRINCIPLE	31
	8.2 OPERATION	31

CHAPTER NO	TITLE	PAGE NO
	8.3 LIMITATION OF THE PRINCIPLE	31
	8.4 PREPARATION AND SETUP	32
	8.5 SENSING AND DATA ACQUISITION	32
	8.6 ELECTRICAL MEASUREMENT	32
	8.7 ENVIRONMENTAL MEASUREMENT	32
	8.8 DATA TRANSMISSION AND DASHBOARD	32
	8.9 DIGITAL TWIN COMPARISON	32
	8.10 FAULT HANDLING AND SHUTDOWN	33
9	**CONCLUSION**	34
10	**FUTURE SCOPES**	37
11	**REFERENCES**	37

LIST OF FIGURES

CHAPTER	FIGURE NO	TITLE	PAGE NO
7	7.1.1	SOLAR PANEL	26
	7.2.1	ESP32 DEVKIT V1	27
	7.3.1	INA219 CURRENT SENSOR	28
	7.4.1	DC VOLTAGE SENSOR	29
	7.5.1	DHT11 SENSOR	30
	7.6.1	LDR LIGHT SENSOR	31

LIST OF TABLES

S.NO	TABLE NO	TITLE	PAGE NO
1		PROGRAM OUTCOMES	viii
2		TABLE OF CONTENT	xi
3		LIST OF FIGURES	xv
4		LIST OF TABLES	xvi
5		LIST OF ABBREVIATIONS	xvii

LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	WORD
1	ADC	ANALOGUE-TO-DIGITAL CONVERTER
2	AI	ARTIFICIAL INTELLIGENCE
3	AO	ANALOGUE OUTPUT
4	BLE	BLUETOOTH LOW ENERGY
5	CE	COMMUNITY EDITION
6	CNN	CONVOLUTIONAL NEURAL NETWORK
7	DC	DIRECT CURRENT
8	DHT11	DIGITAL HUMIDITY AND TEMPERATURE SENSOR
9	DKASC	DESERT KNOWLEDGE AUSTRALIA SOLAR CENTRE
10	DO	DIGITAL OUTPUT
11	GPIO	GENERAL-PURPOSE INPUT/OUTPUT
12	I2C	INTER-INTEGRATED CIRCUIT
13	IEC	INTERNATIONAL ELECTROTECHNICAL COMMISSION
14	INA219	HIGH-SIDE CURRENT AND POWER MONITOR
15	IoT	INTERNET OF THINGS
16	LDR	LIGHT DEPENDENT RESISTOR
17	LM393	DUAL VOLTAGE COMPARATOR
18	MCMC	MARKOV CHAIN MONTE CARLO
19	NREL	NATIONAL RENEWABLE ENERGY LABORATORY
20	NTC	NEGATIVE TEMPERATURE COEFFICIENT
21	PV	PHOTOVOLTAIC
22	PVDAQ	PHOTOVOLTAIC DATA ACQUISITION
23	SCL	SERIAL CLOCK
24	SDA	SERIAL DATA
25	SDG	SUSTAINABLE DEVELOPMENT GOAL
26	SPI	SERIAL PERIPHERAL INTERFACE
27	SRAM	STATIC RANDOM-ACCESS MEMORY

S.NO	ABBREVIATION	WORD
28	UART	UNIVERSAL ASYNCHRONOUS RECEIVER-TRANSMITTER
29	USB	UNIVERSAL SERIAL BUS

CHAPTER-1
INTRODUCTION

1.1 General Introduction

Solar photovoltaic (PV) generation has become one of the main sources of new electrical capacity worldwide. Utility-scale solar farms now contain thousands of modules, hundreds of strings and many inverters, and they are built with the expectation of operating for twenty-five years or more. The financial case for such a plant rest on the energy it delivers over this whole period. The output of a PV plant does not stay constant, however. Modules age, and their ability to convert sunlight into electricity falls slowly with time and exposure.

Degradation has many causes. Cell and interconnect ageing raises the series resistance of the module. Moisture ingress, cell damage and potential-induced effects can lower its shunt resistance. Discolouration of the encapsulant and loss of cell quality reduce the photocurrent. Dust, soiling and partial shading add further temporary losses, and high operating temperature speeds up most ageing processes. These mechanisms act together, and each of them lowers the same measured quantity, which is the output power of the plant.

A digital twin is a virtual model of a physical asset that is kept in step with the asset through measured data. In a PV plant, the twin combines a physical description of the modules with the irradiance, temperature and electrical data recorded in the field. It can then be used to follow the health of the plant, to separate a real fall in performance from a change in weather, and to support decisions about cleaning, repair and replacement.

The project, titled “PV - EVOTWIN - A Self-Evolving Multi-Physics Digital Twin for Utility-Scale Solar PV Farms”, develops such a twin. It is described as multi-physics because it links the electrical behaviour of the modules with their thermal behaviour and with the incident irradiance, and it uses these together to explain the measured output. It is described as self-evolving because the twin does not remain fixed after it is built. Its internal parameters are updated as new field data arrive, so that the model follows the plant as the plant ages.

The performance of any such twin depends on a basic question: what can actually be learned about the degradation state of a plant from the data that the plant records? Answering this question honestly is the central theme of the work.

1.2 Need for the Project

Operators of large solar farms need to know more than whether the output has fallen. They need to know why it has fallen. A fall caused by soiling is corrected by cleaning, a fall caused by a failed string needs repair, and a gradual loss caused by module ageing may need a warranty claim or a planned replacement. Treating one cause as another wastes money and leaves the real problem in place.

Diagnosis in the field is difficult because the available data are sparse. Most plants record irradiance, module temperature and electrical output at the level of the string or the inverter, at fixed time intervals. They do not record the full current–voltage curve of every module. Laboratory methods that rely on dense and precise measurements cannot be applied directly to this kind of data.

A common response is to build a model of the module, with several physical parameters, and to fit it to the field data. When many parameters are fitted together from few measurements, the problem becomes ill-posed. Different combinations of parameters can produce almost the same output, so the data cannot tell them apart. A fitting routine will still return a single set of numbers, and these numbers look precise although they are not reliable. An operator who acts on such numbers may be misled about the true condition of the plant.

A further need comes from the nature of a real plant. A twin that is calibrated once and then left unchanged becomes less accurate as the plant ages, because the plant itself is changing. A useful twin must therefore be able to update itself from new data, and it must do so in a way that remains consistent with the physics of the modules.

There is also a practical need for tools that use open and widely available data. Public monitoring datasets from operating PV systems make it possible to develop and test methods without access to a private plant, and they allow the results to be examined by others. These needs together define the purpose of the project.

1.3 Importance of the Work

The importance of this work lies mainly in its honesty about uncertainty. Many estimation methods return a single best value for each parameter and say nothing about how far it can be trusted. The method proposed here is identifiability-aware. It first examines which parameters, or which combinations of parameters, the available data can determine. It then

estimates those with a Bayesian procedure, which gives a probability distribution for each parameter and not a single number. A narrow distribution shows that the data identify the parameter well. A wide distribution shows that they do not, and the operator is told not to rely on that value.

This approach has several practical benefits. It prevents false precision, because a parameter that cannot be determined is reported as uncertain. It keeps the estimates inside physical limits through prior knowledge of the modules. It allows the same framework to work with different amounts of data, since the reported uncertainty narrows as more informative data become available. It also gives maintenance planners a clearer basis for action, because they can see how confident the twin is in each of its conclusions.

The work also has a wider importance for the solar industry. As more capacity is installed, the cost of operating and maintaining plants becomes a large part of their lifetime cost, and small improvements in diagnosis can lead to large savings across many plants. A twin that can be built from routine field data, with open-source software, can be adopted by smaller operators who cannot afford specialized monitoring equipment. This matters especially in regions where the number of utility-scale plants is growing quickly.

Finally, the reframing of the original problem is itself a contribution. The first concept was to estimate six independent material properties jointly from sparse field sensors. A critical technical review showed that the sensors could not carry enough independent information for this, and the problem was restated as an identifiability-aware estimation of coupled parameters. This change makes the project more rigorous and more defensible, both as an engineering method and as a subject for a patent and a research paper.

1.4 Overview of the Proposed Project

PV - EVOTWIN is a software-based digital twin that estimates and tracks the degradation state of a utility-scale PV farm from routine field data. The twin is built in layers, so that each part can be developed and tested separately.

The data layer provides the plant measurements, which include irradiance, module temperature and electrical output. For development and validation, these data come from two public collections, the NREL PVDAQ datasets and the DKASC datasets, both of which contain measurements from operating PV systems. The device and data-handling layer uses Things Board CE to receive, store and organize the telemetry. The model layer contains the physical

description of the modules and the estimator. The service layer is a Fast API backend, written in Python, which exposes the estimator and its results. The presentation layer is a Grafana dashboard, which shows the health of the plant and the degradation estimates together with their uncertainty.

At the centre of the model layer is the identifiability-aware Bayesian estimator. It works in three steps. It first checks the sensitivity of the measured output to each parameter, in order to find what the data can determine. It then estimates the coupled parameters jointly, so that the dependence between them is preserved, and uses physical limits as priors. Finally, it reports the full distribution of each parameter. The self-evolving behaviour comes from repeating this process as new data arrive, so that the estimates are revised and the model follows the ageing of the plant.

The twin combines a physical model with data-driven updating. The physical model provides the structure and keeps the results consistent with the electrical and thermal behaviour of the modules. The data adjust the parameters of that structure. This combination reduces the amount of data needed and avoids results that break physical laws, which suits the sparse measurements available in the field.

The development followed a compressed execution plan that covered the backend, the Bayesian estimation, the dashboard and the preparation of a paper. The work was supported by a literature review of 36 annotated references and by an analysis of prior art. The method has also been documented for a provisional patent filing.

1.5 Scope of the Project

The scope of the project covers the physical modelling of the PV module and plant, the design of the identifiability-aware Bayesian estimator, the software implementation of the twin, validation on public datasets, and the building of a dashboard for the plant operator. It also covers the documentation of the method as a patent application and a research paper.

The study is carried out on data and software. It does not include the manufacture of any physical hardware, and it does not involve changing the plant in any way. The twin observes the plant and reports on it, and the decisions that follow from its results remain with the operator.

The first phase of the work establishes the framework. It defines the problem, reviews the literature, sets out the architecture and describes the estimation method. Later phases extend the

validation, test the twin on data from partner plants, and improve its coverage of effects such as soiling and shading.

The project does not claim that every degradation parameter can be recovered from every dataset. It states which parameters the data support and which they do not, and it reports the uncertainty in each case. This limitation is part of the method and is discussed throughout the report.

CHAPTER-2
LITERATURE REVIEW

[1] Jordan and Kurtz (2013): Reviewed degradation rates of flat-plate modules and systems from forty years of field testing, assembling nearly 2000 rates with a median of about 0.5 % per year. Outcome: Established the range of degradation that a PV twin must be able to track and showed that reported rates vary widely with technology and climate.

[2] Jordan et al. (2016): Compiled a compendium of published PV degradation rates, covering about 11,000 rates from 40 countries. Outcome: Provided a reference set against which the degradation estimated by the twin can be compared for plausibility.

[3] Jordan et al. (2017): Reviewed the failure and degradation modes reported for modules in the field and the way they change with technology and climate. The work notes that several modes often act together, which makes the power impact of a single mode hard to isolate. Outcome: Supported the view that degradation parameters are coupled and cannot be treated as independent, which is the basis of the re framed problem statement.

[4] Cronin et al. (2014): Presented a method to find PV degradation rates without irradiance data, using relative yield comparisons between systems and a Bayesian analysis to convert relative rates to absolute rates. Outcome: Showed that Bayesian reasoning can extract degradation information from limited field data, many plants having no irradiance sensor.

[5] Jordan et al. (2018): Proposed a robust year-over-year method that uses modeled clear-sky irradiance instead of site sensors and gave the lowest uncertainty among the compared methods on a fleet of 486 PV systems. Outcome: Demonstrated that sensor drift, soiling and data shifts bias degradation estimates, so a twin must treat data quality and uncertainty explicitly.

[6] Zaharatos et al. (2014): Built a statistical model for current–voltage–irradiance data using the five-parameter single-diode model, found maximum-likelihood estimates with confidence intervals, and examined identifiable and stickability. Outcome: Gave a direct precedent for testing which single-diode parameters the data can determine before estimating them.

[7] Zaharatos, Tenorio and Campanelli (2015): Studied the estimability of the single-diode model parameters, using data cloning as one of the tools. Outcome: Confirmed that some

parameter combinations are poorly determined by measured data, which supports the identifiability-aware design of the estimator.

[8] Ayang et al. (2019): Combined a maximum-likelihood estimator with Newton–Raphson solution to identify the unknown parameters of a single-diode module model at different test conditions, and found that accuracy depends on the initial parameters. Outcome: Showed the strength and the limits of point-estimate fitting, which motivates reporting full parameter distributions.

[9] Faiman (2008): Showed that a modified Hottel-Whillier-Bliss equation predicts PV module temperature within the cell-to-cell differences usually found in a module, with constants that did not depend on module type for the modules studied. Outcome: Provided the thermal sub-model that links irradiance, ambient temperature and wind to module temperature in the multi-physics twin.

[10] Holmgren, Hansen and Mikofski (2018): Described pvlib python, an open-source package for modelling solar energy systems. A later update describes more than 100 empirical and physics-based models, including solar position, irradiance, thermal and PV electrical models. Outcome: Supplied tested, open implementations of the physical models used as the backbone of the twin.

[11] Tao et al. (2019): Reviewed digital twins in industry, covering their key components, development and applications, and described the seamless integration of cyber and physical spaces. Outcome: Provided the definition and architecture of a digital twin on which the project is built.

[12] Yuan et al. (2024): Reviewed digital twins and data fusion for PV management and classified PV models as digital models, digital shadows and digital twins according to their data connection. The review also notes that fully integrated twins need advanced analytics, high computing cost and strong security. Outcome: Clarified what separates a true twin from a monitoring model and identified the open challenges that the project addresses.

[13] Olayiwola et al. (2025): Gave a system-level overview of digital twins for renewable energy with a focus on PV, and observed that implementation in PV remains limited compared with wind energy. Outcome: Established that PV digital twins are an immature field, which supports the novelty of the proposed work.

[14] Abdelrahman et al. (2024): Proposed a digital-twin approach that compares array currents computed by the twin model with observed output to detect and classify events in PV arrays. Outcome: Showed the residual-based use of a twin for fault detection, which differs from estimating the degradation parameters themselves.

[15] Fadel and Alelaj (2025): Developed a digital twin for a 250 W PV system that combines a physics-based model with AI prediction and adjusts itself through self-correcting algorithms. Outcome: Gave a small-scale example of a hybrid, self-updating twin, whereas the present work addresses utility-scale plants with sparse data and uncertainty.

[16] Kapteyn, Pretorius and Willcox (2021): Proposed a probabilistic graphical model in which the physical asset and its digital twin are coupled dynamical systems, with the twin assimilating observational data to keep its models in step with the asset. Outcome: Provided the mathematical basis for a twin that evolves as the plant ages.

[17] Kennedy and O'Hagan (2001): Presented a Bayesian technique for calibrating computer models in which the predictions allow for the remaining uncertainty over the fitted parameters. Outcome: Supported the use of Bayesian calibration, in place of ad hoc fitting, to tune the physical model to field data.

[18] Raissi, Perdikaris and Karniadakis (2019): Introduced physics-informed neural networks as a deep learning framework for forward and inverse problems that involve nonlinear partial differential equations. Outcome: Showed how physical laws can be built into data-driven models, which guided the physics-informed part of the twin.

[19] Deline et al. (2021): Described the NREL PVDAQ as a large time-series database of system metadata and performance data used for performance and degradation analysis. Some systems also include environmental sensors for irradiance, temperature, wind speed and precipitation. Outcome: Identified an open, citable dataset suitable for validating the estimator.

[20] Desert Knowledge Australia Solar Centre (DKASC): A demonstration site operating since 2008 that covers many solar technologies, ages, makes, models and configurations, and offers its data as open access. Outcome: Provided a second, long-running dataset from a different climate, so the method is not validated on one source alone.

[21] Laevens et al. (2023): Estimated the azimuth, tilt and rated power of 12 PV systems from openly available PVOutput data, comparing observed clear-day profiles with clear-sky profiles

modelled in PVLib. The fits used Markov Chain Monte Carlo (emcee), so each parameter comes with an uncertainty from the posterior. Outcome: Showed a sampling-based Bayesian workflow that uses pvlib and sparse open data, which supports reporting parameter distributions rather than point estimates.

[22] Jordan and Kurtz (2013): Reviewed field-measured degradation rates of flat-plate modules and systems from about 40 years of literature, gathering nearly 2000 rates with a median of 0.5%/year. Outcome: Gave the baseline range of real-world degradation against which the twin's estimated degradation parameters can be checked.

[23] Jordan et al. (2016): Re-examined published degradation data, now more than 11,000 rates from almost 200 studies in 40 countries. Median degradation for crystalline-silicon technologies was 0.5–0.6%/year, and degradation rates referenced to nameplate rating can be distorted by beginning-of-life deviations and measurement uncertainty. Outcome: Updated the benchmark for degradation rates by technology and warned that the reference baseline matters, which motivates estimating degradation from the plant's own data.

[24] Skomedal and Deceglie (2020): Proposed the combined estimation of degradation and soiling (CODS) method, which extracts degradation, soiling and seasonality from the daily-aggregated normalized performance signal in one self-consistent fit. It is implemented in the open-source RdTools library. Outcome: Showed how to separate degradation from soiling and seasonal effects, which is a key identifiability problem for a utility-scale twin.

[25] Zgraggen et al. (2022): Proposed physics-informed deep learning for tracker fault detection in utility-scale PV plants. A physical model generates faulty data from healthy field data to train a convolutional neural network, and the approach outperformed a purely data-driven one under diverse weather conditions. Outcome: Gave an example of combining a physical model with machine learning when labelled fault data are scarce, which supports the physics-informed part of the twin.

CHAPTER-3
BACKGROUND STUDY

3.1 Photovoltaic Effect and Module Structure

A photovoltaic (PV) cell is a semiconductor junction that converts light directly into electrical energy. When photons with enough energy strike the cell, they generate electron–hole pairs. The built-in electric field of the p–n junction separates these charge carriers, and a current flows through an external circuit connected to the cell. The current is nearly proportional to the incident irradiance, while the voltage depends on the junction properties and on the temperature of the cell.

A commercial module is made by connecting many cells in series, so that the output voltage is useful, and by laminating them between a glass front sheet, polymer encapsulant layers and a back sheet. Modules are then connected into strings, strings into arrays, and arrays to inverters that feed the grid. A utility-scale farm therefore contains thousands of modules, and its output is the combined result of their electrical, thermal and optical behaviour. Every layer of this structure, including the cells, the solder bonds, the encapsulant and the junction box, can change as the module ages.

3.2 Equivalent-Circuit (Single-Diode) Model

The electrical behaviour of a cell or module is commonly described by the single-diode equivalent circuit. It consists of a current source that represents the photocurrent, a diode that represents the junction, a series resistance and a shunt resistance. The terminal current is given by

$$I = I_{ph} - I_0[\exp((V + I R_s) / (n N_s V_T)) - 1] - (V + I R_s) / R_{sh} \quad (1)$$

where I_{ph} is the photocurrent, I_0 is the diode saturation current, R_s and R_{sh} are the series and shunt resistances, n is the diode ideality factor, N_s is the number of cells in series and V_T is the thermal voltage. These five quantities, I_{ph}, I_0, R_s, R_{sh} and n, are the parameters of the model. Because they carry physical meaning, their values act as indicators of the health of the module. A rise in R_s points to interconnect and solder-bond ageing, a fall in R_{sh} points to leakage paths, and a fall in I_{ph} points to optical losses such as discolouration or soiling.

The same model can be used in two directions. In the forward direction, known parameters, irradiance and temperature give the expected current and power. In the inverse direction, measured data are used to find the parameters. The inverse direction is the one needed for degradation estimation, and it is also the more difficult one, as discussed in Section 3.7.

3.3 Effect of Irradiance and Temperature

Module output changes continuously with the weather. The photocurrent rises almost linearly with irradiance, so the power of a module follows the sunlight during the day. Temperature has the opposite effect on voltage. As the cell temperature rises, the open-circuit voltage falls, and the power of a crystalline silicon module typically falls by a few tenths of a percent for every degree Celsius. Module ratings are therefore given at standard test conditions, with an irradiance of 1000 W/m^2 and a cell temperature of 25 °C, while field conditions are almost always different.

For degradation studies this has an important consequence. A drop in power observed in the field may come from weather, from temperature or from a real change in the module. To identify degradation, the weather and temperature effects must first be modelled and removed, so that what remains can be attributed to the module. A twin that models these effects explicitly is better placed to do this than a method that works on the power signal alone.

3.4 Thermal Model of the Module

Module temperature is not measured on every module, and it depends on the irradiance, the ambient temperature and the wind. A simple and widely used description is the Faiman model [9], which is a modified form of the Hottel-Whillier-Bliss heat-balance equation used for flat-plate solar collectors. The module temperature is given by

$$T_m = T_a + G / (U_0 + U_1 v) \qquad (2)$$

where T_a is the ambient temperature, G is the plane-of-array irradiance, v is the wind speed, and U_0 and U_1 are heat-loss coefficients. The model has been adopted in the IEC 61853 standards, and an open-source implementation is available in the pvlib library [10]. Linking this thermal equation to the electrical model of Section 3.2 is what makes the twin multi-physics. The electrical model predicts the power, the power produces heat, and the thermal model returns the temperature that changes the electrical parameters again.

3.5 Degradation Mechanisms and Rates

Degradation is the slow loss of module performance with time and exposure. Reviews of field data report a median degradation rate of about half a percent per year for flat-plate modules, with wide variation between technologies, climates and installations [1]. A later compendium collected many thousands of reported rates from dozens of countries [2]. Rates are not fixed. They depend on temperature, humidity, ultraviolet exposure, mechanical stress and system voltage.

Several mechanisms contribute. Corrosion and fatigue of solder bonds and ribbons raise the series resistance. Moisture ingress, cell cracking and potential-induced degradation can lower the shunt resistance. Encapsulant discolouration and delamination reduce the light reaching the cells and so lower the photocurrent. Reviews of failure and degradation modes note that several of these acts together in the same module, which makes the effect of any one mode on power difficult to separate [3]. This coupling is central to the project. Since the parameters change together, they should be estimated together, and the data should be tested for their ability to tell them apart.

3.6 Sparse Field Data and Data Quality

Operating plants record limited data. Typical records contain irradiance, module or ambient temperature, wind speed and electrical output at string or inverter level, stored at fixed intervals of a few minutes. They do not contain the complete current–voltage curve of each module. Two open collections show this kind of data. The NREL PVDAQ database holds time-series performance data and system metadata from many sites, and most of its data are stored at fifteen-minute intervals [19]. The Desert Knowledge Australia Solar Centre has been operating since 2008 and provides open data from many module technologies, ages and configurations [20].

Field data also contain errors. Sensors drift, inverters clip power at high irradiance, soiling builds up and is washed away by rain, and outages leave gaps. Studies on degradation-rate determination show that such effects can bias the result, and that methods using year-over-year comparison and modelled clear-sky irradiance reduce this bias [5]. Statistical approaches have also been used to estimate degradation when irradiance sensors are not available [4]. For a digital twin, data qualification is therefore a necessary first stage before any parameter is estimated.

3.7 The Inverse Problem and Identifiability

Estimating degradation parameters from measured output is an inverse problem. It is well-posed if one set of parameters produces the observed data and no other set does. It is ill-posed if different sets give almost the same data. In the single-diode model, for example, a change in one resistance can be offset by a change in another parameter over the range of operating conditions that the field data cover, so the data cannot separate them.

This property is called identifiability. A parameter or a combination of parameters is identifiable if the data determine it uniquely, and it is estimable in practice if the data are informative enough to determine it with useful precision. Statistical studies of the single-diode

model have examined both properties for the five parameters [6], [7].

Their results show that some parameters are poorly determined even when the model is correct. Forcing a fit then gives numbers that look precise but are not reliable. This understanding led to the reformulation of the project. Instead of estimating six independent material properties from sparse sensors, the work estimates coupled parameters and states which of them the data support.

3.8 Bayesian Estimation

Bayesian inference provides a natural framework for the inverse problem. The unknown parameters θ are treated as random quantities. Prior knowledge, such as physical limits on resistances, is expressed as a prior distribution p(θ). The measured data d are linked to the parameters through a likelihood p(d | θ), which includes the measurement noise. Bayes' theorem combines them into the posterior distribution

$$p(\theta \mid d) \propto p(d \mid \theta)\, p(\theta) \quad (3)$$

The posterior describes everything that is known about the parameters after the data have been seen. A narrow posterior shows a well-determined parameter. A broad posterior shows that the data say little about it, and the width itself is the measure of uncertainty. For models of this kind the posterior is usually computed by sampling methods such as Markov chain Monte Carlo, since it cannot be written in closed form.

Bayesian calibration of a computer model, set out by Kennedy and O'Hagan [17], is the general method of tuning a physical model to observations while keeping all remaining uncertainty in the predictions. In the present project the single-diode and thermal models play the role of the computer model, and the field data play the role of the observations. When an identifiability check is added, parameters that the data cannot determine are reported with wide posteriors and are not forced to a single value.

3.9 Digital Twin Concepts

A digital twin is a virtual representation of a physical asset that is connected to it through data and used for monitoring, analysis and decision support [11]. Reviews of digital twins for PV separate three levels according to the data connection: a digital model has no automatic data flow, a digital shadow receives data from the plant, and a digital twin exchanges information in both directions and is used to act on the plant [12]. The same reviews report that implementing full twins needs advanced analytics, high computing effort and secure data handling, and that applications in PV remain limited compared with other energy fields such as wind [13].

Most existing PV twins compare the output predicted by a model with the measured output, and a large residual is taken as a sign of a fault [14]. Hybrid twins, which combine a physical model with a learned component, have also been demonstrated on small systems [15].

3.10 Self-Evolving Updating

A plant changes during its life, so a twin calibrated once becomes less accurate over time. A self-evolving twin avoids this by updating its model whenever new data arrive. Work on predictive digital twins describes the physical asset and its twin as two coupled dynamical systems, in which the twin assimilates observations to keep its internal models consistent with the changing asset [16].

3.11 Software and Data Platform

The twin is implemented in Python, which gives access to open scientific libraries and to pvlib, whose reference implementations of irradiance, thermal and electrical models serve as the physical backbone [10]. A FastAPI backend exposes the estimator as a service. ThingsBoard CE receives and stores the telemetry, and Grafana presents the plant health and the parameter estimates with their uncertainty. For development and validation, the public PVDAQ and DKASC datasets provide measured data from operating systems without the need for access to a private plant.

CHAPTER-4
PROBLEM IDENTIFICATION

Utility-scale solar PV farms are expected to operate for twenty-five years or more, yet their performance declines steadily from the day of commissioning. Operators usually see only the result, a fall in energy yield, and cannot easily tell which physical mechanism is responsible. The following problems establish the need for the proposed system.

Degradation in a PV farm is not a single process. Soiling, thermal ageing, encapsulant discolouration, interconnect fatigue and cell-level losses all act together, and each changes the electrical output in a similar way. A monitoring system should therefore be judged by how correctly it separates these causes, and not only by how closely it reproduces the measured power.

4.1 Lack of Insight into Degradation Mechanisms

Present monitoring tools report power, performance ratio and alarms, but they do not recover the physical parameters that cause the losses.

A performance ratio that falls by a small amount each year gives no information on whether the cause is dust, module ageing or a failing connection. The corrective action differs for each, since cleaning, replacement and re-wiring have very different costs. Without a physical explanation of the loss, maintenance decisions are made on schedule or on experience, and not on evidence.

4.2 Static Models That Do Not Evolve

Conventional simulation models use fixed parameters taken from the datasheet and become inaccurate as the plant ages. A model calibrated once at commissioning describes a new plant and not an ageing one. As modules degrade, the gap between predicted and measured output grows, and the model can no longer serve as a reliable reference for fault detection or yield forecasting. A model that is updated continuously from field data is needed so that the digital representation follows the real plant through its life.

4.3 Sparse Sensing and the Ill-Posed Inverse Problem

Field plants measure only a few quantities, such as irradiance, module temperature and string power, while the number of coupled degradation parameters to be recovered is larger.

Estimating several material and loss parameters at the same time from a small number of measurements is an ill-posed inverse problem. Different combinations of parameters can produce nearly the same output, so a simple curve-fitting method may return a confident but wrong answer. This is the central technical difficulty of the work, and a method that ignores it will give misleading diagnoses.

4.4 Absence of Uncertainty Information

Point estimates of degradation are of limited use unless they are accompanied by a statement of how reliable they are.

An operator deciding whether to replace a string of modules needs to know how certain the estimate is. A deterministic fit gives a single number and hides the fact that some parameters are well determined by the data and others are not. A probabilistic treatment, which reports a distribution and identifies the parameters that the data cannot resolve, allows decisions to be made with a clear understanding of risk.

4.5 Disconnected Physics and Data-Driven Methods

Purely physical models are rigid, while purely data-driven models need large datasets and give results that are difficult to interpret or trust.

Machine-learning models can follow the measured output closely, but they do not respect physical laws and may behave unpredictably outside the range of their training data. Physical models respect the laws but cannot capture every effect seen in the field. Most available tools use one approach or the other, and few combine the two within a framework that is both accurate and explainable.

4.6 Problem Statement

Utility-scale PV farms need a way to identify the causes of performance loss from the limited data they collect, but existing tools are static, report only aggregate losses, and give no measure of confidence. A self-updating, physics-based digital twin that estimates coupled degradation parameters with stated uncertainty is therefore needed.

The proposed system therefore sets out to provide a multi-physics digital twin that is updated from field data, an identifiability-aware Bayesian estimator that recovers degradation

parameters and reports which of them can actually be determined, a software platform that collects the data and presents the results on a dashboard, and validation against public field datasets. The system is intended for utility-scale plants with the sensing normally available, and its limits will be stated clearly after validation.

CHAPTER-5
IMPACT OF THE PROBLEM

5.1 Loss of Energy Yield

If degradation is not identified and corrected in time, the plant produces less energy than it could for the rest of its life.

A small annual loss may seem unimportant, but it accumulates over the life of the plant and applies to the whole installed capacity. Losses from causes that can be corrected, such as soiling or a failing string, continue unnoticed when they are mixed with unavoidable ageing in a single performance figure. Separating the two allows the recoverable part of the loss to be addressed.

5.2 Economic Loss and Investment Risk

Lost energy is lost revenue, and uncertainty about the true condition of a plant affects its value and financing.

The return on a solar project depends on the energy it delivers over many years. Lenders, insurers and buyers need a reliable estimate of the remaining performance, and a plant whose degradation is poorly understood carries a higher risk premium. Inefficient maintenance adds to the cost, since crews may be sent to clean or replace components that are not the real cause of the loss.

5.3 Inefficient Operation and Maintenance

Maintenance planned without knowledge of the cause of loss is either too frequent or too late.

Fixed-interval cleaning and inspection are simple to arrange but wasteful, since the condition of the plant varies from site to site and from season to season. Diagnosis based on estimated parameters would allow work to be directed to the strings and mechanisms that need it. The benefit depends on the quality of the estimates, which is why the treatment of uncertainty is an important part of the proposed method.;

5.4 Wrong Diagnosis from Ill-Posed Estimation

Estimators that ignore identifiability can attribute a loss to the wrong mechanism with apparent confidence.

If two causes produce almost the same change in output, a conventional fit will choose one of them arbitrarily. The operator then acts on a diagnosis that the data did not support. An identifiability-aware method reduces this risk by stating when a parameter cannot be separated from another, so that the result is reported as uncertain and not as a false certainty.

5.5 Reliability and Grid Considerations

Large solar plants contribute a growing share of grid supply, so inaccurate knowledge of their condition affects forecasting and scheduling.

Forecasts of plant output are used by grid operators to balance supply and demand. A forecast based on a model that has drifted away from the real plant will be less accurate, and the error must be covered by other sources. A digital twin that tracks the plant as it ages would improve the quality of such forecasts, but it cannot remove the variability caused by weather, and it should be regarded as an aid to planning and not a guarantee of output.

5.6 Environmental and Sustainability Impact

Solar plants are installed to reduce emissions, and any avoidable loss of output reduces that benefit.

Every unit of energy lost to undetected degradation must be supplied from another source, which may be a fossil fuel. Longer useful life and better maintenance also reduce the need for early module replacement, and so reduce the material and energy cost of manufacture and disposal. The overall effect will depend on how widely such tools are adopted and on the practices of the plant operator.

CHAPTER-6
OBJECTIVES

The overall objective is to design, develop and validate a self-evolving multi-physics digital twin for utility-scale solar PV farms that estimates coupled degradation parameters from sparse field data with a measure of uncertainty.

The proposed system combines a physics-based model of the PV plant, an identifiability-aware Bayesian estimator, a data acquisition and storage layer, a backend service and a dashboard. Its performance will be evaluated through the accuracy of parameter recovery, the quality of the uncertainty estimates, the agreement between predicted and measured power, and the response time of the software.

6.1 Primary Objectives

- To develop a multi-physics digital twin that represents the electrical, thermal and degradation behaviour of a PV plant.
- To estimate coupled degradation parameters from sparse sensor data using a Bayesian method.
- To make the estimator aware of identifiability, so that unresolvable parameters are reported as such.
- To update the model continuously from field data so that it follows the ageing plant.
- To present the results on a dashboard that is useful to plant operators.

The first objective is to build a digital twin whose structure follows the physical behaviour of the plant. The model should relate irradiance and temperature to electrical output and should include the degradation mechanisms that change this relationship over time. It should be detailed enough to explain the main losses, yet simple enough to be estimated from the sensors normally installed.

The second objective is to recover the degradation parameters with a Bayesian estimator. The method should return a distribution for each parameter and not a single value, so that the user can see how well the data determine it. Prior knowledge from the physics of the problem should be used to constrain the estimates where the data are weak.

The third objective is to make the estimation identifiability-aware. The estimator should detect which parameters or combinations of parameters cannot be separated with the available measurements and should report them as uncertain. This is the main contribution of the work, because it addresses the ill-posed nature of the inverse problem directly.

The last primary objective is to make the twin useful in practice. The model should be updated as new data arrive, and the dashboard should show the estimated parameters, their uncertainty and the difference between predicted and measured output, so that an operator can use the information to plan maintenance.

6.2 Technical Objectives

- To formulate the forward model linking irradiance, temperature and degradation parameters to power output.
- To implement the Bayesian estimator in Python and test it first on synthetic data with known parameters.
- To validate the method on the NREL PVDAQ and DKASC field datasets.
- To build the data pipeline and dashboard using ThingsBoard CE and Grafana.
- To develop a FastAPI backend that connects the data, the estimator and the dashboard.
- To measure parameter recovery error, prediction error and computation time.
- To document the limits of the method, including the conditions under which parameters cannot be identified.

The first technical objective is to formulate the forward model. The model should be derived from the physics of the module and the plant, and its parameters should be defined clearly so that each can be linked to a physical degradation mechanism. The assumptions made should be stated and checked against the data.

The second technical objective is to implement the estimator and test it on synthetic data. Since the true parameters are known in a synthetic case, the recovery error and the behaviour of the identifiability analysis can be measured directly before the method is applied to field data, where the true values are not known.

The third objective is to validate the method on public field datasets. The results should be compared with measured output, and the stability of the estimates over time and between sites

should be examined. Cases in which the method performs poorly should be reported and not omitted.

The fourth and fifth objectives concern the software platform. Data should be collected and stored through ThingsBoard CE, processed through a FastAPI backend that calls the estimator, and displayed in Grafana. The interfaces between the parts should be defined clearly so that each can be tested and replaced independently.

The final objective concerns performance evaluation. Testing should record the error in recovered parameters on synthetic data, the error between predicted and measured power on field data, the coverage of the stated uncertainty ranges, and the time taken to update the model. The conditions under which the method fails to resolve the parameters should be recorded and stated with the results.

CHAPTER-7
HARDWARE REQUIREMENTS

7.1 Solar Photovoltaic Panel

The solar panel is the energy source of the system and the device whose behaviour the digital twin represents. It is made of silicon photovoltaic cells connected in series to raise the output voltage. When sunlight falls on a cell, photons with sufficient energy lift electrons across the semiconductor junction. The resulting separation of charge produces a potential difference, and a direct current flows through any load connected across the terminals. The electrical output depends mainly on the light intensity and on the cell temperature. The current rises almost in proportion to irradiance, while the voltage falls slowly as the cell temperature rises. This is why temperature and light measurements are taken together with the electrical readings.

A small-format panel is used in this work because it is safe to handle and convenient to mount on a laboratory test bench. The rated voltage and current printed on the back of the panel set the limits of the measuring circuits. The voltage divider and the current sensor must both be able to handle the open-circuit voltage and the short-circuit current of the panel. The panel should be mounted at a fixed tilt, facing the sun, with the light sensor and the temperature sensor placed beside it so that they experience the same conditions. The wiring should be kept short, and the polarity must be checked before connection.



Fig 7.1.1 – solar panel

7.2 ESP32 DevKit V1 Microcontroller

The ESP32 DevKit V1 is the central unit of the data acquisition system. It is built around the ESP32 system-on-chip, which has a dual-core Xtensa LX6 processor running at up to 240 MHz, 520 KB of SRAM and, on this board, typically 4 MB of flash memory. The chip has built-in Wi-Fi (2.4 GHz, 802.11 b/g/n) and Bluetooth (classic and BLE), so no external communication module is needed. This is the main reason for choosing it over a conventional 8-bit controller, since the measured data must be sent to a dashboard for the digital twin to work. The board works at 3.3 V logic and is powered through its USB port or the VIN pin, and an on-board regulator supplies the 3.3 V rail.

The 30-pin board has a 12-bit analogue-to-digital converter, I2C, SPI and UART interfaces and a large number of general-purpose pins. In this project the I2C bus uses GPIO21 for data and GPIO22 for the clock and connects the current sensor. The analogue outputs of the voltage sensor and the light sensor are connected to input-only pins in the range GPIO34 to GPIO39, which belong to the first ADC unit. This unit is chosen because the second ADC unit cannot be used while Wi-Fi is active. The temperature and humidity sensor is connected to a free digital pin. The ADC of the ESP32 is not perfectly linear, and its input must not exceed 3.3 V, so the readings are calibrated in software and the inputs are protected by the divider ratios. The program reads every sensor at a fixed interval, converts the raw values to engineering units, calculates the power and sends the data to the monitoring dashboard.



Fig 7.2.1 – ESP32 DevKit V1

7.3 INA219 Current and Power Sensor

The INA219 is a high-side current and power monitor with an I2C interface. A precision shunt resistor is connected in series with the load, and the chip measures the small voltage drop across it. The module carries a 0.1 Ω shunt (marked R100), and the current follows from Ohm's law as the measured drop divided by the shunt resistance. The chip also measures the bus voltage, up to 26 V, and with these two values it supplies the power internally. Its internal 12-bit converter gives good resolution for the currents of a small solar panel. It operates from a 3 V to 5.5 V supply, so it can be powered directly from the 3.3 V rail of the ESP32.

The module has pins for supply, ground, clock and data, and two terminals, VIN+ and VIN−, through which the measured current passes. The panel is connected to VIN+ and the load to VIN−, so that the current flows through the shunt. The device address can be changed with the address pins if more than one sensor is on the bus. The shunt adds a small series drop, which is acceptable for a low-power panel. The calibration register of the chip is set in the program to match the actual shunt value and the expected maximum current, otherwise the current and power readings will be scaled wrongly

Features:

- Measures current, bus voltage and power in a single device.
- High-side current sensing using a 0.1 Ω (R100) shunt resistor.
- Bus voltage measurement up to 26 V.
- 12-bit internal ADC for good resolution.



Fig 7.3.1 – INA219 Current Sensor

7.4 DC Voltage Sensor Module (0–25 V)

The voltage sensor module is a resistive potential divider on a small board. It uses a 30 kΩ and a 7.5 kΩ resistor, which gives a division ratio of 5:1. A voltage applied to the input terminals therefore appears at the signal pin as one-fifth of its value, and the program multiplies the measured value by five to recover the actual voltage. The module has three pins: the positive input, the ground and the analogue signal output. The input terminals are on the side that connects to the panel, and the signal pin goes to an ADC input of the ESP32.

The module is rated for inputs up to 25 V on a 5 V controller. The ESP32 ADC accepts only 3.3 V, so the usable input is limited to about 16.5 V, and a margin below this should be kept. This is acceptable for a small panel, whose open-circuit voltage is well below this limit. The divider draws a very small current from the panel, so it does not disturb the measurement. Its accuracy depends on the tolerance of the resistors and on the ADC of the ESP32, so the readings are compared with a multimeter and a correction factor is applied in the program. The panel voltage measured here, together with the current from the INA219.

Features:

- Simple resistive divider using 30 kΩ and 7.5 kΩ resistors.
- Division ratio of 5:1, so the actual voltage is the output multiplied by five.



Fig 7.4.1 – DC Voltage Sensor

7.5 DHT11 Temperature and Humidity Sensor

The DHT11 measures the temperature and relative humidity of the air around the panel. It contains an NTC thermistor for temperature and a capacitive humidity element, whose capacitance changes as the polymer layer absorbs moisture. A small chip inside the sensor converts both readings to a digital signal and sends it through a single data line using a timing-

based protocol. The sensor works from 3.3 V to 5.5 V and measures 0 to 50 °C with an accuracy of ±2 °C, and 20 to 90 % relative humidity with an accuracy of ±5 %. It gives a new reading about once per second, which is sufficient because ambient conditions change slowly.

The module has three pins: supply, ground and data. The data pin is connected to a digital pin of the ESP32, and a pull-up resistor is usually already fitted on the module board. The sensor should be placed in the shade, close to the panel but not touching it, so that it records the air temperature and not the heat of the panel surface. It should also be kept out of direct sunlight, which would raise the reading. Temperature has a clear effect on the voltage and efficiency of a solar panel, and humidity affects the transmission of light and the dust deposited on the surface, so both are inputs of the digital twin model.

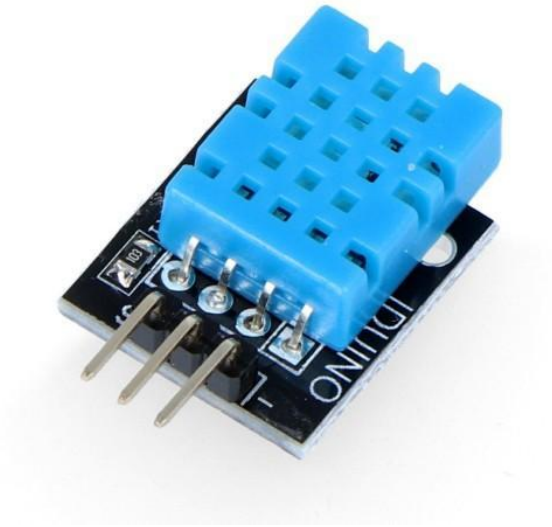


Fig 7.5.1 – DHT11 Sensor

7.6 LDR Light Sensor Module (LM393)

The light sensor module uses a light dependent resistor, whose resistance falls when the light intensity rises and rises again in darkness. The resistance is high in the dark and drops to a low value in strong light. The LDR forms a voltage divider with a fixed resistor, so the light level appears as a voltage on the analogue output pin. The module also has an LM393 comparator, which compares this voltage with a threshold set by an on-board potentiometer and gives a digital high or low output. The module works from 3.3 V to 5 V and has four pins: supply, ground, analogue output and digital output. The sensor is mounted in the same plane as the panel and faces the same direction, so that it receives the same light. The LDR responds differently to

different wavelengths and is not linear, so its reading is relative and does not give irradiance in W/m². The values are used to follow the changes in sunlight, such as passing clouds, and to relate them to the changes in panel power.

Features:

- Light dependent resistor whose resistance falls as light intensity increases.
- LM393 comparator gives a clean digital output.
- Both analogue (AO) and digital (DO) outputs are available.

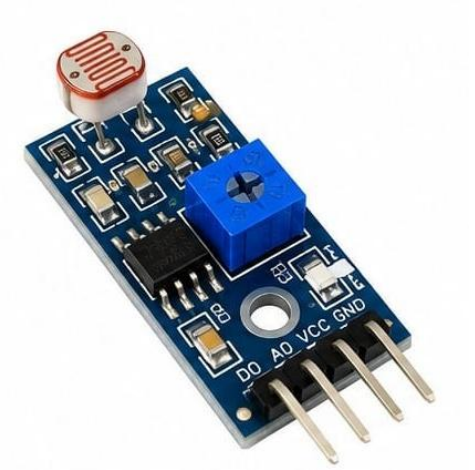


Fig 7.6.1 – LDR Light Sensor

7.7 Interconnection of the Hardware

All the sensors used in the solar photovoltaic monitoring system are interfaced with the ESP32 DevKit V1, which acts as the central data acquisition and processing unit. The ESP32 collects information from the different sensors, processes the measured parameters and transmits the combined data to the IoT dashboard for real-time monitoring and performance analysis.

1. Power Supply and Ground Configuration

The low-voltage sensor modules are supplied through the 3.3 V output pin of the ESP32, provided that their individual operating voltage and current requirements are satisfied. The DHT11, INA219 and LDR sensor modules receive their required supply voltage through the appropriate power connections. The common ground configuration is maintained between the

ESP32, sensor modules and panel measurement circuits to establish a common voltage reference for reliable signal measurement.

A common grounding arrangement is particularly important for the analogue sensors because their output signals are measured with respect to the ESP32 ground reference. Proper grounding helps minimize unwanted voltage variations and improves the consistency of the acquired sensor readings.

2. INA219 Current Sensor Interfacing

The INA219 current sensor is connected to the ESP32 through the Inter-Integrated Circuit (I2C) communication interface. Its Serial Data (SDA) and Serial Clock (SCL) pins are connected to the corresponding GPIO21 and GPIO22 pins of the ESP32, respectively.

The solar panel supplies electrical power to an external load through the INA219 module's shunt resistor, which is connected in series with the load circuit. By measuring the voltage drop across this shunt resistor, the sensor determines the current flowing through the circuit. It also measures the bus voltage and calculates the corresponding electrical power.

The measured voltage, current and power values are transferred digitally to the ESP32 through the I2C communication protocol. This arrangement enables continuous monitoring of the electrical output and operating performance of the photovoltaic panel.

3. DC Voltage Sensor Interfacing

The DC voltage sensor module (0–25 V) is connected in parallel across the positive and negative terminals of the solar panel. It operates using a resistive voltage-divider circuit, which reduces the input voltage to a lower level suitable for measurement by the ESP32's analogue-to-digital converter (ADC).

The reduced voltage signal is connected to an appropriate analogue input pin of the ESP32. The controller reads this signal and calculates the actual solar panel voltage using the voltage-divider conversion ratio.

This arrangement allows the system to continuously monitor variations in the panel's output voltage under different environmental and operating conditions. The sensor output must remain within the ESP32's permissible ADC input range, which is approximately 0-3.3 V.

4. DHT11 Temperature and Humidity Sensor Interfacing

The DHT11 sensor is connected to the ESP32 through a dedicated digital GPIO pin. It measures the surrounding temperature and relative humidity using its internal sensing elements and converts the measured values into digital data.

The ESP32 communicates with the DHT11 using its single-wire digital communication protocol. The collected temperature and humidity readings are periodically transferred to the controller for further processing.

These environmental parameters are useful for analyzing the operating conditions of the solar photovoltaic system. In particular, temperature variations can influence photovoltaic module efficiency and electrical power generation, while humidity provides additional information about the surrounding environmental conditions.

5. LDR Light Sensor Interfacing

The LDR light sensor module, equipped with an LM393 comparator, is connected to the ESP32 to monitor changes in surrounding light intensity. The resistance of the light-dependent resistor varies according to the amount of incident light, producing a corresponding change in its output voltage.

The analogue output (AO) of the module is connected to an ESP32 ADC input to obtain continuously varying sensor readings. The digital output (DO), if used, provides a HIGH or LOW signal based on the light threshold adjusted through the onboard potentiometer.

6. Signal Conditioning and Noise Reduction

Proper wiring and signal management are essential for obtaining accurate sensor measurements. The connections between the sensors and ESP32 are maintained as short as practically possible to reduce electromagnetic interference and unwanted signal variations.

Where necessary, twisted-pair or shielded cables are used for sensitive analogue signal connections. The power and ground wiring is arranged carefully to minimize voltage fluctuations caused by the operation of the external load. Suitable decoupling capacitors and stable power connections can also be employed to reduce electrical noise and improve measurement reliability. These precautions are particularly important for analogue voltage and light sensor readings, where electrical interference may introduce errors into the acquired data.

7. Data Acquisition and Digital Twin Integration

The ESP32 continuously collects data from all the connected sensors and combines the measured parameters into a single monitoring dataset. The acquired information includes solar panel voltage, current, electrical power, ambient temperature, relative humidity and light intensity. After processing, the ESP32 transmits the sensor readings through its built-in Wi-Fi communication capability to the IoT platform and monitoring dashboard.

The collected real-time data are then compared with the corresponding outputs generated by the digital twin model. This comparison helps evaluate the performance of the photovoltaic system, identify deviations between predicted and actual operating conditions, and analyse the influence of environmental variations on power generation. Through this integrated arrangement, the ESP32 serves as the communication link between the physical photovoltaic monitoring setup and its virtual digital twin representation, enabling continuous data monitoring, performance evaluation and future predictive analysis.

CHAPTER-8
WORKING PRINCIPLE

8.1 Operating Principle

The system works on the photovoltaic effect combined with real-time sensing and a digital twin. When sunlight falls on the silicon cells of the solar panel, photons transfer their energy to electrons in the semiconductor junction. This creates a potential difference across the panel terminals, and a direct current flows through any load connected to it. The power produced depends mainly on the light intensity and on the temperature of the cells. The current rises almost in proportion to the irradiance, and the voltage falls slowly as the cell temperature rises.

A digital twin is a virtual model that follows the behaviour of the physical panel. To keep the model in step with the real panel, the actual operating conditions must be measured continuously. In this system, one group of sensors measures the electrical output of the panel (voltage, current and power), and another group measures the conditions that affect it (light intensity, air temperature and humidity). The ESP32 collects all these readings, converts them to engineering units and sends them over Wi-Fi to the monitoring dashboard. There the measured power is compared with the power predicted by the model for the same conditions. A difference between the two shows that the panel is not behaving as expected, which may be due to dust, shading, ageing, loose connections or a fault.

8.2 Operation

1. The solar panel receives sunlight and generates DC electrical power.
2. The panel current passes through the shunt resistor of the INA219, which measures the current, the bus voltage and the power.
3. The DC voltage sensor, connected across the panel terminals, gives an analogue voltage equal to one-fifth of the panel voltage.
4. The LDR module gives an analogue voltage that shows the relative intensity of the sunlight falling on the panel.
5. The DHT11 measures the temperature and the relative humidity of the air near the panel and sends them as a digital signal.
6. The ESP32 reads the INA219 over the I2C bus, the voltage and light sensors through its ADC inputs, and the DHT11 through a digital pin.
7. The program converts the raw readings to volts, amperes, watts, degrees Celsius and percentage humidity, applies the calibration factors and calculates the power.
8. The data are transmitted through Wi-Fi to the dashboard at fixed intervals, where they are displayed and compared with the digital twin output.

For example, when a cloud passes over the panel, the LDR reading falls first, followed by the current and power of the panel. The dashboard records this change, and the digital twin predicts a matching fall in power. If the measured fall is greater than the predicted one, the system shows that some other loss is present.

8.3 Limitation of the Principle

The main disadvantage of the system is that the accuracy of the digital twin depends on the accuracy of the measured inputs. The LDR gives only a relative light reading and not the irradiance in W/m², and the DHT11 has a limited accuracy of ±2 °C. These limits restrict how closely the model prediction can be matched with the real output, so the system is best suited to tracking trends and detecting deviations.

8.4 Preparation and Setup

Before operation, the panel surface is inspected and cleaned, and the wiring and connectors are checked for firm connection and correct polarity. The rated voltage and current of the panel are noted from its label to confirm that they lie within the range of the voltage sensor and the INA219. The ground of all the modules is joined together, and the output of the voltage sensor is checked to ensure that it stays below 3.3 V at the highest panel voltage.

The panel is placed at a fixed tilt facing the sun. The LDR is mounted in the same plane as the panel, and the DHT11 is placed in the shade close to the panel but not touching it. The ESP32 is then powered through its USB port, and its connection to the Wi-Fi network and to the dashboard is verified. A few readings of the voltage and current are compared with a multimeter, and the correction factors are entered in the program before the trial begins.

8.5 Sensing and Data Acquisition

The ESP32 acts as the data acquisition unit of the system. The program runs in a continuous loop. In each cycle it reads every sensor, converts the values, calculates the power and sends the data. The cycle is repeated at a fixed interval, which is long enough for the slow DHT11 to give a fresh reading and short enough to follow rapid changes in sunlight.

The analogue signals are sampled by the 12-bit ADC. Because a single sample is affected by noise, several samples are taken in quick succession and averaged. The readings of the voltage sensor and the LDR are connected to the first ADC unit of the ESP32, since the second unit cannot be used while Wi-Fi is active. The I2C bus carries the INA219 readings on two wires, and the DHT11 data line is read using its timing-based protocol.

8.6 Electrical Measurement

The INA219 is connected in series with the panel output. The load current flows through

its 0.1 Ω shunt resistor and produces a small voltage drop proportional to the current. The chip measures this drop and find the current from Ohm's law:

$$I = \frac{V_{shunt}}{R_{shunt}}$$

The bus voltage is measured at the same time, and the chip supplies the power internally. The program also calculates the power from the product of the voltage and current as a check:

$$P = V \times I$$

The DC voltage sensor gives a second, independent measurement of the panel voltage. Its two resistors divide the input in the ratio 5:1, so the program recovers the actual voltage by multiplying the measured value by five:

$$V_{in} = V_{out} \times 5$$

Comparing the voltage from the divider with the bus voltage of the INA219 confirms that both measuring paths agree and helps to reveal any wiring or calibration error.

8.7 Environmental Measurement

The LDR and its resistor form a voltage divider. In bright light the resistance of the LDR is low and the output voltage changes in one direction. In dim light the resistance is high and the voltage changes in the opposite direction. The ESP32 reads this voltage and converts it to a relative light intensity. The LM393 comparator on the module also gives a digital signal when the light crosses the threshold set by the potentiometer. This can be used to indicate the start and end of the day or heavy cloud.

The DHT11 measures the air temperature with its thermistor and the relative humidity with its capacitive element. Temperature affects the voltage and efficiency of the panel, and humidity affects the light reaching the cells and the dust settling on the surface. These three environmental readings are the inputs that the digital twin model needs to calculate the expected output of the panel.

8.8 Data Transmission and Dashboard

After each cycle, the ESP32 sends the voltage, current, power, temperature, humidity and light intensity to the dashboard through its Wi-Fi connection. The values are stored with the time of measurement so that the history of the panel can be plotted. The dashboard displays the present readings and the graphs of the power, voltage and current against time, and shows the environmental conditions alongside them.

If the Wi-Fi connection is lost, the ESP32 keeps reading the sensors and tries to reconnect, and the transmission resumes when the link is restored. This prevents the loss of the measurement process because of a short network failure.

8.9 Digital Twin Comparison

The digital twin takes the measured light intensity and temperature as its inputs and calculates the power that the panel should produce under those conditions. The calculation follows the usual behaviour of a photovoltaic panel, in which the output increases with irradiance and decreases as the cell temperature rises above its reference value. The predicted power is then compared with the power measured by the INA219.

When the two values agree within a small tolerance, the panel is considered healthy. When the measured power is persistently lower than the predicted power, the difference indicates a loss caused by factors such as dust, partial shading, a loose connection or ageing of the cells. The difference between predicted and measured values is displayed on the dashboard, and its trend over several days shows whether the panel is degrading slowly or has developed a sudden fault. The comparison also makes it possible to adjust the model parameters so that the digital twin follows the real panel more closely.

8.10 Fault Handling and Shutdown

The program checks every reading for a valid range. A DHT11 reading that fails, a voltage sensor output at the limit of the ADC, or a current of zero during bright sunlight is treated as a sensor or wiring fault and shown on the dashboard as an error message. A reading outside the permitted range is not used in the calculation, so that a single bad value does not distort the comparison with the model. The voltage sensor input must never be allowed to exceed the safe limit of the ESP32, and the circuit is disconnected if the panel voltage rises beyond it.

For shutdown, the last readings are sent to the dashboard, the load is disconnected from the INA219 and then the power to the ESP32 is removed. The operator records the date and time, the highest power reached, the light and temperature conditions and any faults observed in the trial.

These steps provide a repeatable operating sequence for monitoring the solar panel and comparing its measured performance with the digital twin.

CHAPTER-9
CONCLUSION

The project on the solar photovoltaic digital twin provides a low-cost method of monitoring a solar panel and comparing its measured behaviour with a virtual model. An ESP32 microcontroller collects the data from an INA219 current sensor, a DC voltage sensor, a DHT11 temperature and humidity sensor and an LDR light sensor. The readings are converted to engineering units, sent over Wi-Fi to a dashboard and compared with the power predicted by the digital twin. The system therefore shows not only how much power the panel produces but also whether it is producing the power that it should under the existing light and temperature conditions.

The main limitations come from the sensors. The LDR gives only a relative light reading and not the irradiance in W/m², and the DHT11 has a limited accuracy of ±2 °C. These limits restrict how closely the model can be matched with the real panel, so the present system is best suited to tracking trends and detecting deviations rather than to precise laboratory measurement. With measured data collected over several days, the model parameters can be adjusted and the prototype refined for practical use.

The design combines a few simple and widely available parts in a way that can be built, understood and repaired by a student or a small workshop. The solar panel is the source of power, the sensors measure the electrical output and the operating conditions, the ESP32 acquires and transmits the data, and the dashboard and the model give the meaning to the numbers.

The data acquisition is simple but effective for this application. The use of two independent measurements of the panel voltage, one from the INA219 and one from the voltage divider, allows the readings to be cross-checked and helps to reveal wiring or calibration errors. The range checks in the program prevent a single bad reading from distorting the comparison, and the dashboard gives the user the information needed to follow the condition of the panel. More advanced models and sensors can be added later without changing the basic hardware.

The project also shows how an engineering concept is turned into a working system: the literature is studied, the problem is defined, the components are selected, the prototype is built and its performance is tested.

CHAPTER-10
FUTURE SCOPES

The most direct way to improve the accuracy of the system is to replace the LDR with a calibrated irradiance sensor and the DHT11 with a more accurate temperature sensor. A temperature sensor fixed to the back of the panel would measure the cell temperature directly, which is the quantity that the model actually needs. A better ADC, such as an external 16-bit converter, would also reduce the error of the analogue readings.

The digital twin itself can be improved by fitting the model parameters to the measured data and by using machine learning methods. A trained model can learn the behaviour of the panel under different weather conditions and predict its output more closely than a fixed equation. It can also classify the type of loss, such as dust, partial shading or ageing, from the pattern of the difference between the measured and predicted power.

The system can be made more useful by adding automatic alerts. When the measured power remains below the predicted power for a set time, a message can be sent to a mobile phone through the cloud, and the history stored on the server can be used to plan the cleaning and maintenance of the panel. A mobile application would allow the user to follow the panel from any place.

Energy storage and tracking can also be added. A charge controller and battery would allow the stored energy and the efficiency of the whole system to be monitored, and a sun tracking mechanism would increase the energy collected from the panel during the day. Finally, the system can be extended from a single small panel to a complete array, with one ESP32 node for each string and a common dashboard, so that it can be used in a real solar plant.

Monitoring and communication features can also be added. A wireless or GSM module can send the cabinet temperature and a fault alert to a mobile phone, and a data logger can store the temperature history, which is useful for storing medicines. A humidity sensor and a condensate-drain arrangement would make the unit more reliable in humid climates. Finally, the system can be adapted for cooling the cabin of a parked vehicle or a protected area, with the unit switching on automatically when the interior temperature exceeds a set limit.

CHAPTER-11

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